

Net731 Week 3: Bongani Thengeni

20240038

Phase 1: Pre-Deployment Planning

Before touching the portal, define your networking schema to avoid the "default" values prohibited by the task.

- **Region:** Select one close to you (*South Africa North*).
- **Resource Group:** Create a new one (RG-Techbridge)
- **Networking Plan:**
 - **VNet Address Space:** 10.10.0.0/16
 - **Subnet:** 10.10.1.0/

Phase 2: Create the Virtual Network

Do this **before** creating the VM to ensure you are not using auto-generated defaults.

1. Search for **Virtual Networks** > **Create**.
2. **Basics:** Select your Resource Group and Name (Techbridge-RG)
3. **IP Addresses:**
 - Delete the default 10.0.0.0/16.
 - Add your custom space: 10.10.0.0/16.
 - Add a subnet: 10.10.1.0/24.
4. Review + Create.

Microsoft Azure

Home

Create a virtual machine

Help me create a low cost VM

+2

X

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription * ⓘ Azure subscription 1

Resource group * ⓘ Techbridge-RG

Create new

Instance details

Virtual machine name * ⓘ Techbridge-VM

Region * ⓘ (US) Central US

Deploy to an Azure Extended Zone

Availability options ⓘ Availability zone

Zone options ⓘ

☒ Self-selected zone

Choose up to 3 availability zones, one VM per zone

☐ Azure-selected zone (Preview)

Let Azure assign the best zone for your needs

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Review + create

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Upgrade

Search resources, services, and docs (G+7)

Copilot

bthengeni0404@gmail...

DEFAULT DIRECTORY (BTHENG...

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Techbridge-VNet01-1778776475179 | Overview

X

Deployment

Search

Delete Cancel Redeploy Download Refresh

Overview

Inputs

Outputs

Template

Deployment is in progress

Deployment name : Techbridge-VNet01-1778776475... Start time : 5/14/2026, 6:35:16 PM

Subscription : Azure subscription 1 Correlation ID : 58c68637-6970-4038-bcb1-d6d1...

Resource group : Techbridge-RG

Deployment details

Resource	Type	Status	Operation de
Techbridge-VN...	Virtual network	Created	Operation det

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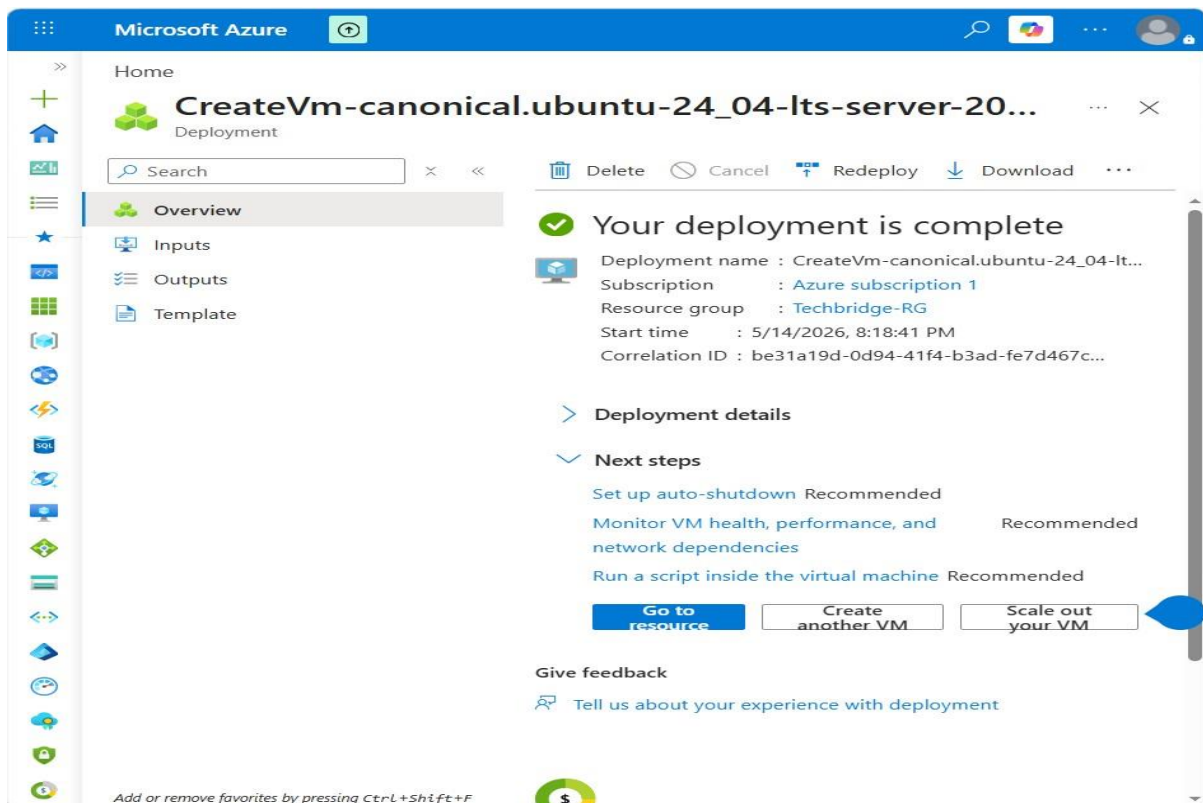
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Phase 3: Deploy the B1s Virtual Machine

1. Search for **Virtual Machines** > **Create** > **Azure Virtual Machine**.
2. **Basics Tab:**
 - **VM Name:** vm-linux-01.
 - **Image:** Ubuntu Server 22.04 LTS.
 - **Size:** Change size to **Standard_B1s** (Filter by name if you don't see it).
 - **Authentication:** Choose **SSH Public Key** (Recommended for Linux/Production) or **Password**. *SSH is more secure against brute-force attacks.*



3. **Networking Tab:**
 - **Virtual Network:** Select the one you created in Phase 2.
 - **Public IP:** Click **Create new**.
 - **CRITICAL:** Change SKU to **Standard** and Assignment to **Static**.
 - **NIC Network Security Group:** Select **Basic**.

- **Public inbound ports:** Allow **SSH (22)** for Linux or **RDP (3389)** for Windows.

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Create a virtual machine

Help me create a low cost VM +2

Project details
Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription * ⓘ Azure subscription 1

Resource group * ⓘ Techbridge-RG
[Create new](#)

Instance details

Virtual machine name * ⓘ Techbridge-VM ✓

Region * ⓘ (US) Central US
[Deploy to an Azure Extended Zone](#)

Availability options ⓘ Availability zone

Zone options ⓘ

☒ Self-selected zone
Choose up to 3 availability zones, one VM per zone

☐ Azure-selected zone (Preview)
Let Azure assign the best zone for your needs

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[Give feedback](#)

4. Management Tab:

- **Auto-shutdown:** Check **Enable auto-shutdown**.
- **Shutdown time:** Set it to a time you are definitely not working (7:00 PM).
- **Time zone:** Ensure that it matches your local time.

5. Review + Create: Click Create once validation passes

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Create a virtual machine

Help me create a low cost VM +2

Microsoft Entra ID login now uses SSH certificate-based authentication. You will need to use an SSH client that supports OpenSSH certificates. You can use Azure CLI or Cloud Shell from the Azure Portal. [Learn more](#)

Auto-shutdown

Enable auto-shutdown ☒

Shutdown time

Time zone

Notification before shutdown ☒

Email *

Backup

Enable backup ☐

Guest OS updates

Enable periodic assessment ☐

Patch orchestration options

Some patch orchestration options are not available for this image. [Learn more](#)

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Review + create

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Extensions + applications

Operating system

Configuration

Advisor recommendations

Properties

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Help me copy this VM in any region +1

Advisor (1 of 4): Enable Trusted Launch foundational excellence, and modern security for Existing Generation 2 VM(s)

Help me copy this VM in any region

Connect Start Restart Stop

Disk

OS disk	Techbridge-VM_OsDisk_1_847516c7155f45efb6c9b54d39828663
Encryption at host	Disabled
Azure disk encryption	Not enabled
Ephemeral OS disk	N/A
Data disks	0

Auto-shutdown

Auto-shutdown	Enabled
Scheduled shutdown	07:00 PM (Coordinated Universal Time)